FINANCIAL MARKETS ANALYTICS

Dynamic Asset Allocation

Variant B

Classical Markowitz | Michaud Resampling
Ledoit-Wolf | Hierarchical Risk Parity

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DETAILED ORAL DEFENSE

REQUIRED VS ADDED

The assignment required dynamic allocation and out-of-sample evidence

REQUIRED CORE

- Dynamic multi-asset allocation
- Classical Markowitz optimization
- Michaud-style resampling
- Walk-forward out-of-sample comparison
- Benchmarks, graphs, and risk metrics

OUR ADDITIONS

- Ledoit-Wolf covariance shrinkage
- Target Volatility portfolio
- Hierarchical Risk Parity extension
- Drift-aware turnover and costs
- Concentration and diversification diagnostics

Variant B: allocation across asset classes through time - not stock selection.

RESEARCH QUESTION

Sophistication must beat transparent rules after costs

Can optimized portfolios improve realized risk-adjusted performance?

01

Markowitz

Does it beat Equal Weight and 60/40?

02

Ledoit-Wolf

Does covariance shrinkage improve stability?

03

Michaud

Does resampling reduce instability?

04

HRP

Does hierarchy offer a robust alternative?

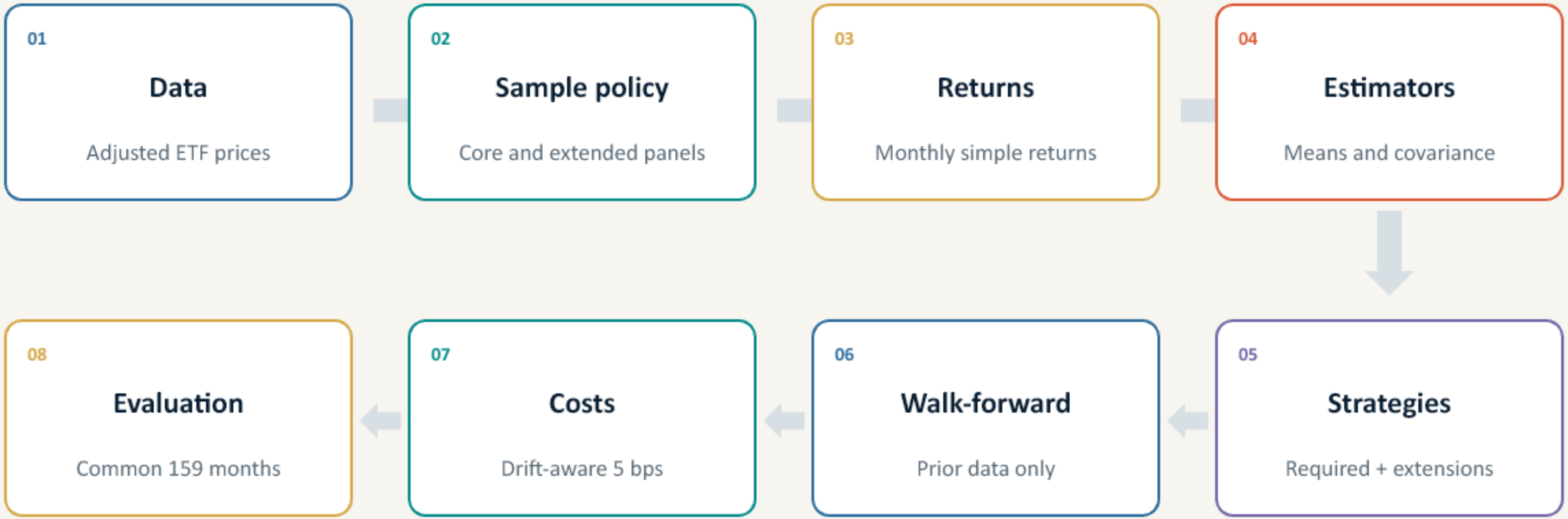
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Implementation

Do gains survive turnover and 5-bps costs?

No strategy receives a more favorable universe, sample, date, or cost convention.

One controlled pipeline separates model choice from sample choice



Required methods and extensions differ only at the strategy stage.

ETFs keep the experiment about allocation, not stock selection

INDIVIDUAL STOCKS

- Firm-specific events dominate outcomes
- Stock-selection noise obscures allocation skill
- Economic interpretation is more difficult
- Cross-asset roles are not represented

MULTI-ASSET ETFs

- Broad, diversified market exposures
- Reduced firm-specific risk
- Liquid and reproducible instruments
- Each weight has a clear asset-class role

One instrument can represent one interpretable source of portfolio risk.

27 CONFIGURED ETFs

The initial 27-ETF universe was broad but still interpretable

27 instruments across six economic sleeves



- Equities: 9
- Government bonds: 7
- Credit: 4
- Commodities: 3
- Real estate: 3
- Cash: 1

WHY THIS BREADTH?

- Growth and geographic equity exposure
- Duration, credit, and inflation sensitivity
- Real assets and cash-like defense
- Within the recommended 15-30 range
- Fixed before strategy performance was compared

Broad enough for allocation; small enough to explain.

ROLE-BASED SELECTION

Each ticker has a distinct economic role

Asset class	ETF examples	Economic purpose
US equities	SPY / QQQ / IWM / MDY	Size and style exposure
International equities	EFA / EEM / VGK / EWJ / VPL	Developed and emerging growth
Government bonds	SHY / IEF / TLT / TIP / BWX	Duration, inflation, global defense
Credit	LQD / HYG / EMB / BKLN	Investment-grade to floating-rate risk
Commodities	GLD / DBC / GSG	Gold and broad real-asset exposure
Real estate	VNQ / RWX / VNQI	US and international listed property
Cash	BIL	Short Treasury-bill exposure

Selection criterion: economic role + liquidity + availability + interpretability - not ex-post performance.

The research universe stayed at 27; the core experiment used 23

27 Initial research universe

FOUR LATE-INCEPTION ETFs

IGOV | VGSH | VNQI | BKLN

No January 2008 history

CORE EXPERIMENT

23 ETFs

from January 2008

EXTENDED PANEL

27 ETFs

from March 2011

No backfilling. No synthetic history. Final optimization uses the core 23 only.

Preserving 2008 adds a severe stress regime to the evidence



- A 2011 start would remove the Global Financial Crisis
- The core choice was fixed before strategy results
- Stress coverage was prioritized over four extra ETFs
- This is not automatic missing-data deletion

Availability determined the universe - not ex-post performance.

YAHOO FINANCE

Adjusted prices support reproducible ETF return histories

WHAT THE PIPELINE USES

- Yahoo Finance accessed through yfinance
- Daily adjusted ETF price histories
- Adjustments reflect distributions and corporate actions
- Daily data converted to calendar month-end prices
- Frozen downloaded files retained for reproducibility

IMPORTANT LIMITATION

Accessible and reproducible

but not equivalent to

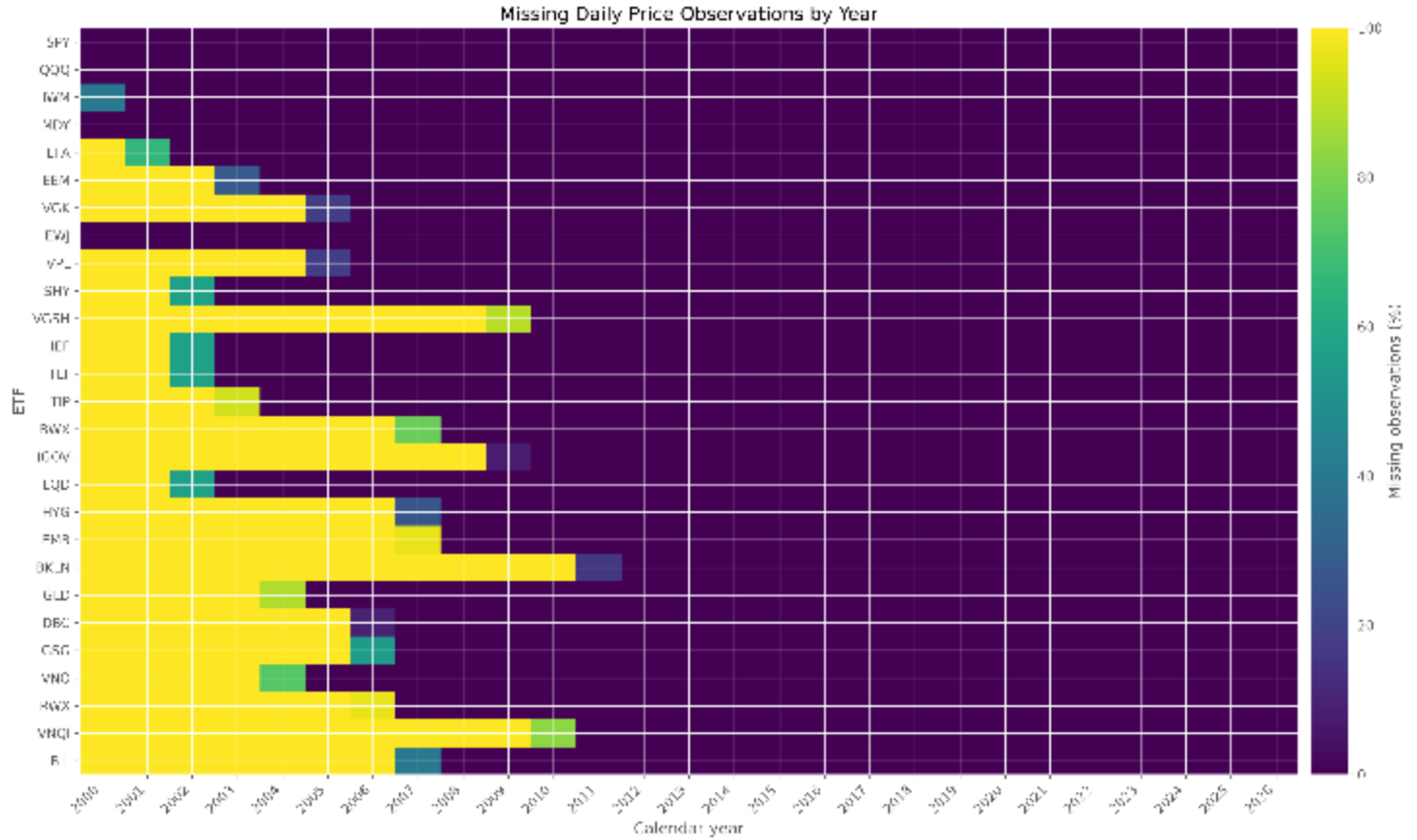
Bloomberg, FactSet, or an institutional total-return database

A new live download may differ if the provider revises adjusted history.

Submitted results use the included frozen raw and processed artifacts.

NO SYNTHETIC DATA

Structural pre-inception gaps were never treated as zero returns



Source: frozen data-quality output

OBSERVED PATTERN

- Missing cells are mainly pre-inception history
- No within-lifetime missing observations

CLEANING RULES

No interpolation
No backfilling
No zero-return substitution
Incomplete June 2026 removed

Unavailable history is not a zero price and not a zero return.

Simple monthly returns drive portfolio wealth

SIMPLE MONTHLY RETURN

$$r(t) = P(t) / P(t-1) - 1$$

Month-end adjusted prices

WHY SIMPLE RETURNS?

- Portfolio return is the weighted sum of asset simple returns
- They compound directly into investor wealth

LOG RETURNS

Retained for diagnostics; not used for final portfolio wealth

GEOMETRIC RETURN

Reported performance reflects realized compounding

SHARPE INPUT

Uses annualized arithmetic mean excess return

Geometric return and the arithmetic mean used in Sharpe are different quantities.

Monthly data balance responsiveness with implementability

DAILY

Too reactive

- More noise
- Unnecessary turnover
- Unstable covariance

MONTHLY

Chosen balance

- Matches assignment
- Practical cadence
- Enough history
- Costs remain visible

ANNUAL

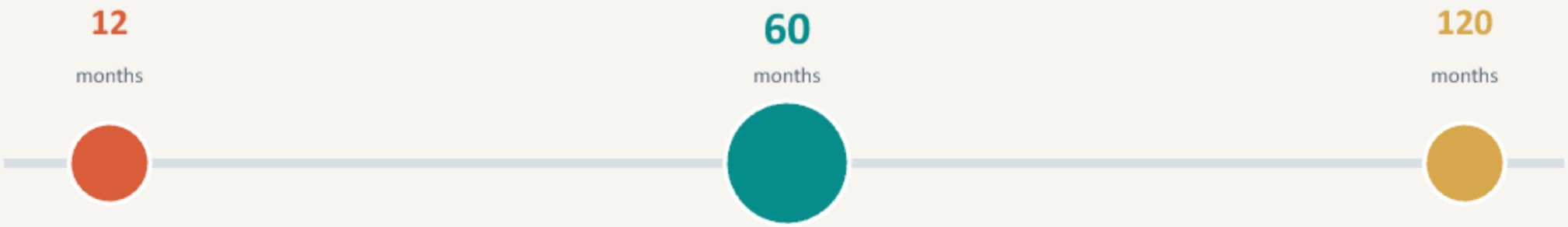
Too slow

- Few observations
- Poor adaptability
- Delayed response

Quarterly rebalancing remains a future robustness test, not part of the frozen comparison.

PROJECT-SPECIFIC CHOICE

A 60-month window balances estimation noise and regime adaptation



Too reactive

Only 12 observations
for 23 assets

Chosen balance

Five years of data
with regime response

Too slow

More stable, but stale
and delays the backtest

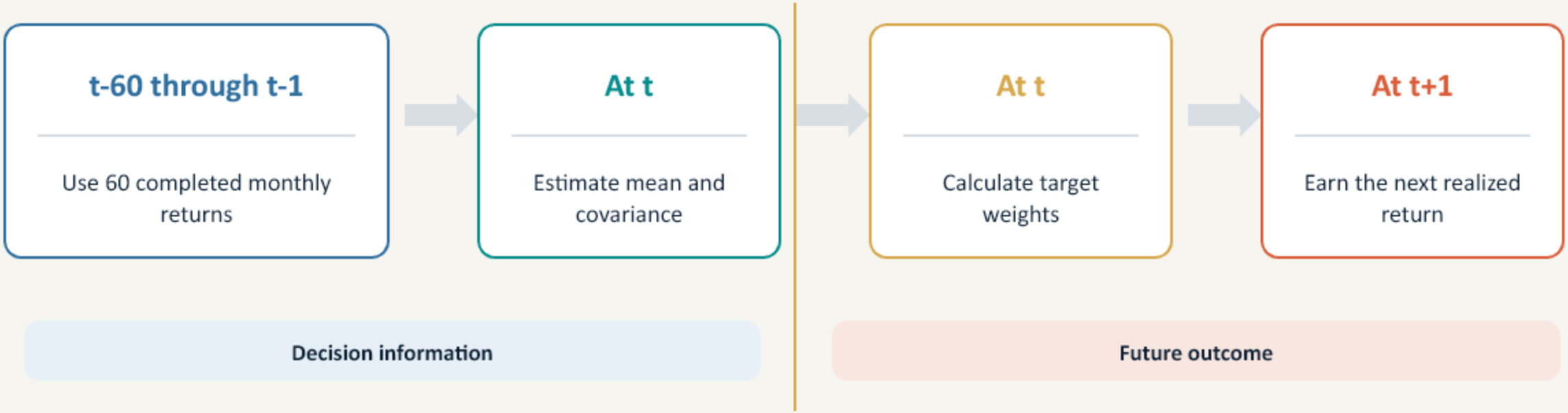
<- ADAPTABILITY INCREASES

STABILITY INCREASES ->

A transparent compromise - not claimed to be universally optimal.

NO LOOK-AHEAD

The realized return never enters its own estimation window



- The estimator ends strictly before the weight date
- The new portfolio is held during the following month

The return earned after the decision cannot help create that decision.

159 REALIZED MONTHS

The first common realized month is March 2013



31 March 2013 -> 31 May 2026

159 identical realized months for every final strategy

February 2013 is a decision date; March 2013 is the first month that earns return.

DRIFT-AWARE TURNOVER

Turnover is measured from drifted holdings to the new target

STARTING TARGET		AFTER MARKET DRIFT		NEW TARGET	
SPY	20%	SPY	23%	SPY	25%
TLT	20%	TLT	18%	TLT	17%

Trade only the differences

SPY: +2 percentage points | TLT: -1 percentage point

Target-to-target changes alone would ignore the return-driven drift between rebalances.

ONE-WAY COST

Five basis points are charged only on the traded fraction

FORMULA

cost = turnover x 0.0005

5 bps = 0.05%

NUMERICAL EXAMPLE

10% one-way turnover

x 0.05% trading cost

= 0.005% of portfolio value

WHAT IS CHARGED

- Only the traded wealth fraction
- Every ongoing rebalance
- Same rule for every final strategy

WHAT IS NOT MODELED

- Taxes and market impact
- Time-varying spreads or liquidity
- Brokerage minimum fees

Initial establishment is excluded from turnover averages.

COMPARABILITY ASSUMPTION

Five basis points provide a modest, consistent implementation penalty

01

Liquid ETF proxy

A modest trading-cost assumption for accessible funds

02

Implementation penalty

High-turnover rules do not trade for free

03

Constant across strategies

Trading intensity is penalized consistently

04

Transparent simplification

Easy to reproduce and audit

Not a universal market estimate: costs vary by ETF, trade size, and market regime.

Future check: time-varying, asset-specific transaction costs.

TRANSPARENT BASELINES

Complexity has value only relative to credible simple benchmarks

EQUAL WEIGHT

- No forecasts
- Maximum capital-weight breadth
- Complete core opportunity set
- Easy to explain and reproduce

60/40

- Traditional balanced allocation
- Growth plus defensive sleeves
- No optimizer or covariance estimate
- Familiar implementation standard

A complex model must improve on rules that have no estimation error.

1/N BENCHMARK

Equal Weight assigns 1/23 to every core ETF

TARGET WEIGHT

1 / 23

approximately 4.35% per ETF

23 effective assets by construction

IMPLEMENTATION

- Every core ETF receives identical target capital
- Monthly drift-aware rebalancing
- Same realized dates and 5-bps cost convention
- No expected-return or covariance estimation
- Equal capital does not mean equal risk contribution

A broad capital allocation benchmark - not a risk-parity portfolio.

TRADITIONAL BALANCED MIX

The 60/40 benchmark uses only equity and government-bond sleeves

60% EQUITIES | **40% BONDS**

9 equity ETFs

60% / 9 = 6.67% each

5 government-bond ETFs

40% / 5 = 8.00% each

Zero benchmark weight

Credit | Commodities | Real estate | Cash

WHY USE IT?

- Familiar policy portfolio
- Clear growth-defense split
- No forecast or optimizer
- Same monthly costs and dates

Limitation: it does not use the complete opportunity set.

INPUTS -> DECISION

Markowitz converts forecasts, covariance, and constraints into weights

EXPECTED RETURNS

mu

Reward estimate

COVARIANCE

Sigma

Risk and co-movement

CONSTRAINTS

C

Investability rules

OUTPUT

w*

Portfolio weights

Minimum Variance

Minimize risk

Maximum Sharpe

Maximize excess return per risk

Target Volatility

Maximize return under a risk ceiling

The framework is disciplined; the inputs remain estimated and noisy.

DEFENSIVE OBJECTIVE

Minimum Variance buys stability, not high expected return

OBJECTIVE

minimize

$$w' \Sigma w$$

Uses covariance + constraints

Does not use expected returns

EXPECTED BEHAVIOR

- Favors low-volatility, low-correlation assets
- Can concentrate in cash and short bonds
- Suitable for capital preservation

PROJECT RESULT

2.25-2.31%	2.77-2.88%	0.096-0.114
Annualized return	Volatility	Sharpe

Low Sharpe: realized return stayed close to the 2% reference rate.

BEST SHARPE: 0.922

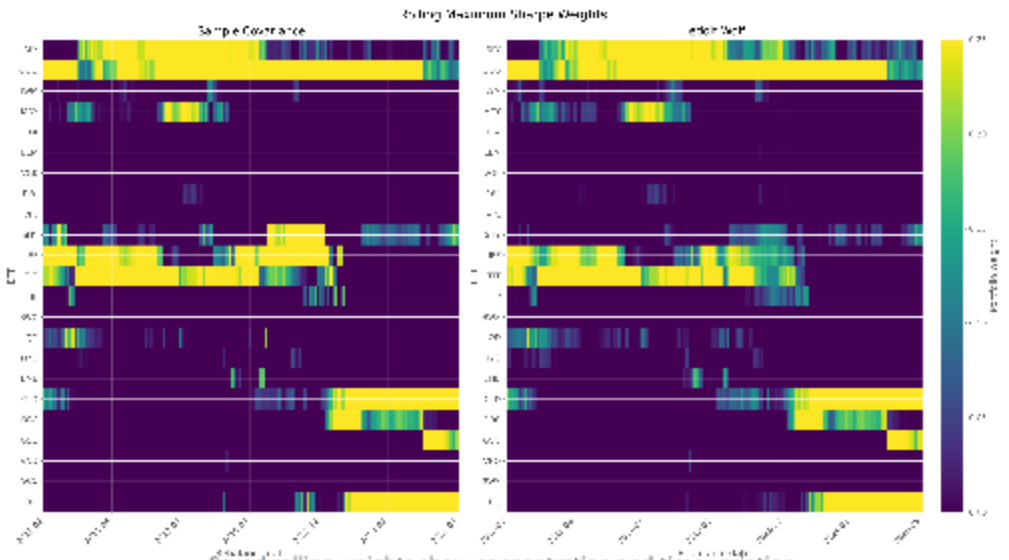
Maximum Sharpe achieved the best realized risk-adjusted result

OBJECTIVE

maximize expected excess return per unit of portfolio risk

Uses mu + Sigma + 2% risk-free rate

- Noisy expected returns
- Concentrated weights
- High rebalancing demand



Saved rolling weights show concentration and time variation.

9.86%	8.45%	88.06%
Annualized return	Volatility	Annualized turnover

Ledoit-Wolf delivered the highest realized Sharpe.

MARKOWITZ OBJECTIVE 3

10% EX-ANTE TARGET

Target Volatility reached the highest wealth by accepting more risk

OBJECTIVE

maximize expected return

subject to

estimated annual volatility <= 10%

Moderate project-specific ceiling

WHY 10%?

- Comparable to balanced portfolios
- Allows meaningful risky-asset participation
- Prevents unconstrained risk escalation

REALIZED RESULT - LEDOIT-WOLF

11.38%	11.63%	4.17
Annualized return	Realized volatility	Final wealth

An ex-ante 10% constraint cannot guarantee 10% realized future volatility.

MARKOWITZ + MICHAUD

Constraints turn mathematical optima into interpretable portfolios

Weights sum to 1	**Fully invested**
Minimum weight 0%	**Long only; no short selling**
Maximum weight 25%	**Limits estimation-driven corner solutions**
No leverage	**Comparable, investable risk**

WHY NOT TIGHTER?

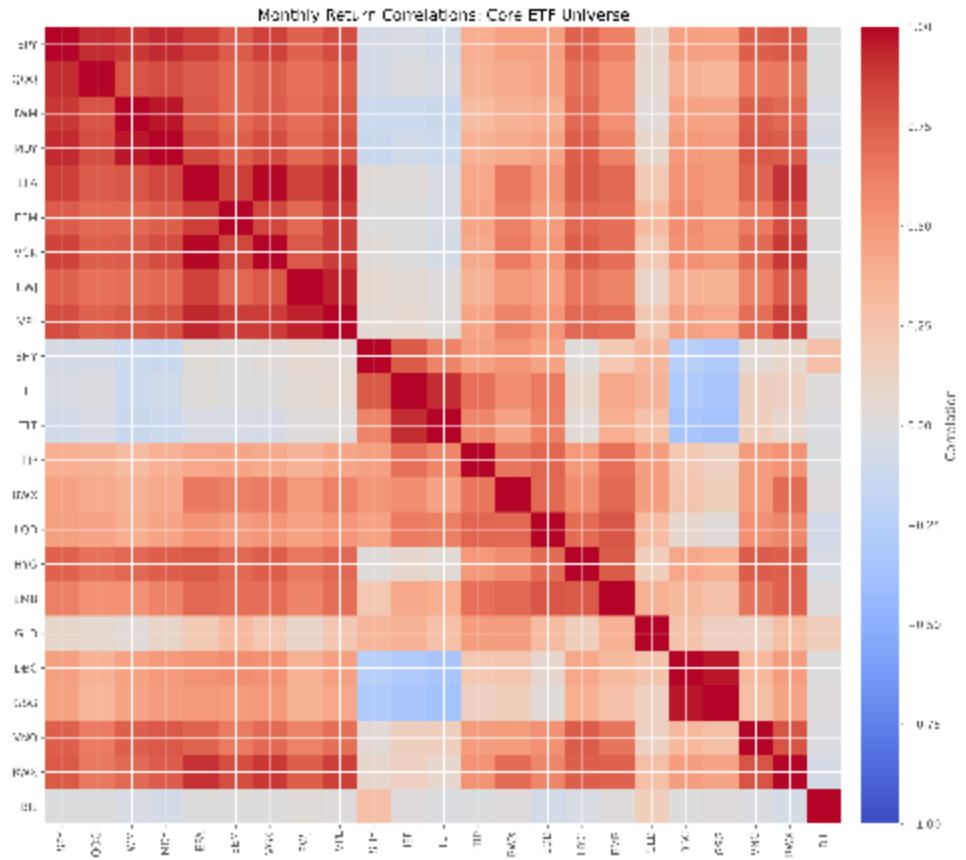
A much lower cap forces quasi-equal weighting.

WHY NOT UNCAPPED?

No cap allows noisy estimates to dominate.

23 x 23 FROM 60 MONTHS

A 23-asset sample covariance is noisy with only 60 observations



Frozen core-universe correlation output

WHY IT MATTERS

- Variances and correlations contain sampling noise
- Optimization amplifies small input errors
- Poor conditioning creates unstable weights
- Equity clustering limits naive diversification

The sample estimator is valid - but high-variance in a short window.

ESTIMATOR - NOT OBJECTIVE

Ledoit-Wolf trades a little bias for much lower estimation variance

SAMPLE COVARIANCE

Detailed but noisy

+

STRUCTURED TARGET

Stable but simplified

→

LEDOIT-WOLF

Weighted combination

Data-driven shrinkage intensity

Small bias → Lower estimation variance → Better conditioning → More stable risk input

The Markowitz objective stays unchanged; only the covariance input is stabilized.

Shrinkage improved the strongest Markowitz implementation

Strategy	Sharpe	Annualized turnover	Effective assets
Min Variance	0.096 -> 0.114	26.06% -> 23.46%	4.71 -> 5.35
Max Sharpe	0.909 -> 0.922	99.14% -> 88.06%	4.50 -> 4.77
Target Volatility	0.822 -> 0.818	103.94% -> 98.66%	4.50 -> 4.52

MAXIMUM SHARPE

Higher Sharpe, lower turnover, slightly broader weights

MINIMUM VARIANCE

Small Sharpe improvement and meaningful turnover reduction

TARGET VOLATILITY

Lower turnover, but no Sharpe improvement

Useful here - not proof of universal dominance.

Michaud was required because one estimated frontier is fragile

01

INPUTS LOOK PRECISE

Estimated means and covariance

02

BUT ARE UNCERTAIN

Sampling error remains

03

OPTIMIZER AMPLIFIES

Favors extreme estimates

04

WEIGHTS BECOME FRAGILE

Concentration and turnover

Michaud response

Average optimized portfolios across plausible capital-market assumptions.

An extension of Markowitz - not an unrelated strategy.

Resampling moderates dependence on one noisy efficient frontier.

SIX-STEP CYCLE

Michaud averages decisions across plausible inputs

01 Estimate
mu and Sigma from prior 60 months

02 Simulate
one plausible market assumption set

03 Optimize
constrained Maximum Sharpe

04 Repeat
M independent scenarios

05 Average
component weights across solutions

06 Hold
averaged portfolio next month

Final weight for asset i

$$w(i) = \text{average of } w(i,1) \ldots w(i,M)$$

The future return is never part of the simulation or estimation window.

Average portfolio decisions - not future outcomes.

NORMAL SAMPLING THEORY

The code resampled capital-market assumptions, not historical rows

WHAT WAS SIMULATED

- Expected-return vectors from their sampling distribution
- Covariance realizations consistent with Sigma and sample size
- One constrained Maximum Sharpe solution per scenario
- Same long-only, no-leverage, 25% cap rules

WHAT IT WAS NOT

- Not a simple bootstrap of historical rows
- Not a simulation of future realized returns
- Not an unconstrained efficient frontier
- Not a claim that returns are perfectly normal

Reproducibility — **Random seed 42 + deterministic seed sequences at each rebalance**

CONVERGENCE CHECK

Three resample counts reveal Monte Carlo convergence

100

resamples

Fastest

More Monte Carlo variation

91.53% turnover

250

resamples

Intermediate

Convergence checkpoint

81.45% turnover

500

resamples

Primary

Results largely stabilized

78.00% turnover

The marginal changes became small; more simulation would add cost, not a new conclusion.

FOCUSED COMPARISON

Michaud sacrificed little Sharpe and improved implementation

Metric	Max Sharpe Ledoit-Wolf	Michaud 500	Interpretation
Return	9.86%	9.23%	Small return sacrifice
Volatility	8.45%	7.98%	Slightly lower realized risk
Sharpe	0.922	0.900	Near-leading risk-adjusted result
Turnover	88.06%	78.00%	Lower implementation burden
Effective assets	4.77	6.25	Broader optimized allocation

Michaud did not beat Markowitz on every metric; it improved robustness and diversification.

ORIGINAL MODERN EXTENSION

OPTIONAL EXTENSION

HRP provides a structurally different robustness test

WHY HRP?

- Avoids direct covariance inversion
- Uses no expected-return forecast
- Clusters related assets hierarchically
- Provides a non-Markowitz comparison
- Tests method-universe interaction

WHY NOT BLACK-LITTERMAN?

- Requires explicit investor views
- Requires confidence levels for those views
- No defensible objective view set was available
- Invented views would add untestable subjectivity
- Reserved for future work

HRP was added to challenge the design - not to guarantee a better result.

SIX-STEP HIERARCHY

HRP allocates through a hierarchy rather than an efficient frontier

01
Distance
Convert correlations to distances

02
Cluster
Group similar assets

03
Reorder
Place related assets together

04
Split
Divide hierarchy recursively

05
Allocate
Weight clusters by variance

06
Continue
Reach individual ETF weights

IT AVOIDS

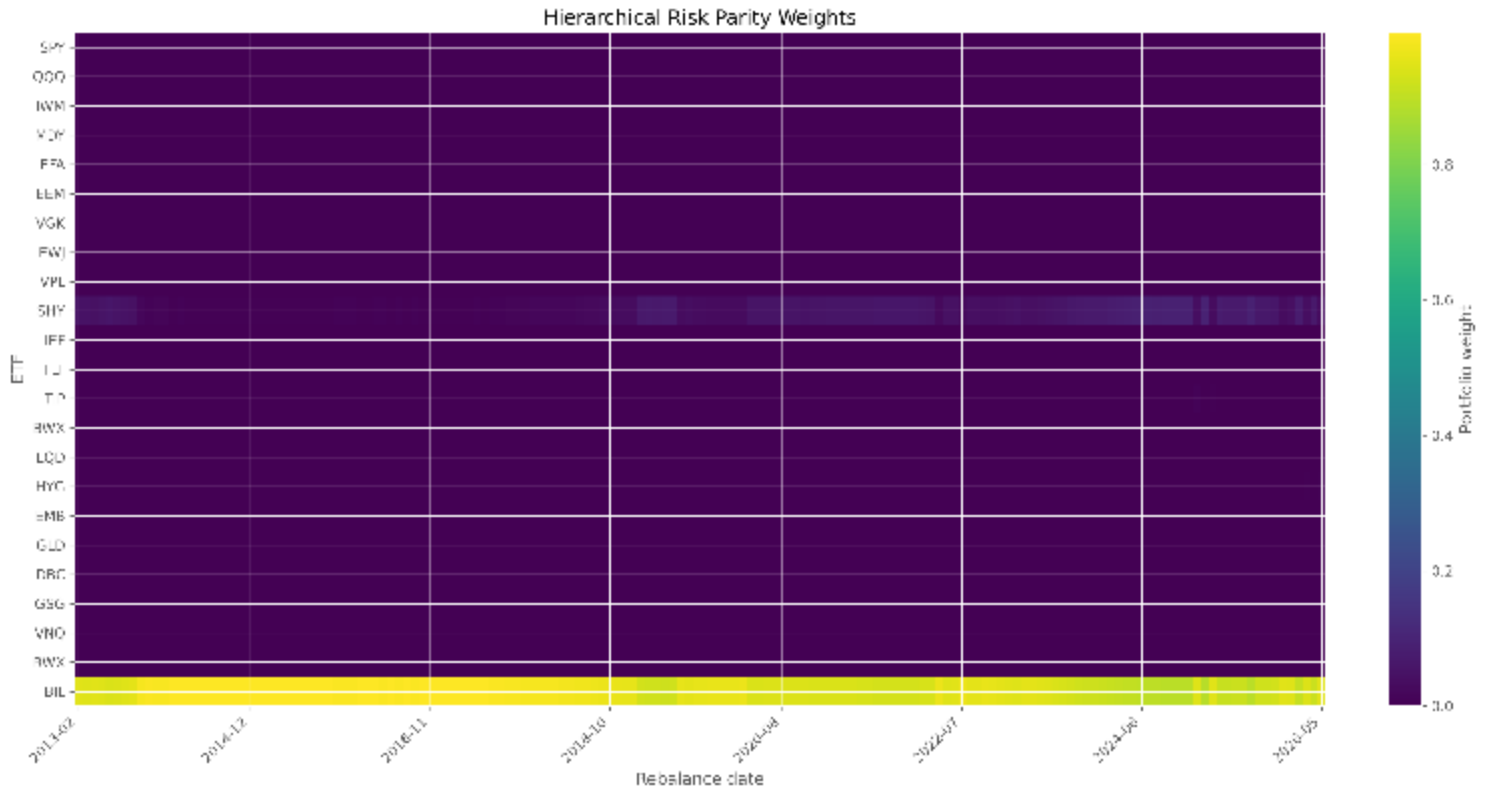
- Expected returns and direct covariance inversion

IT DOES NOT GUARANTEE

- Balanced capital weights or equal risk contributions

Hierarchy changes the allocation logic - it does not remove dependence on the universe.

Unconstrained HRP collapsed toward the cash-like BIL ETF



Saved HRP weights

CAUSAL CHAIN

- BIL has extremely low realized volatility
- Recursive variance allocation strongly favors it
- HRP has no Markowitz-style 25% cap
- Average BIL weight reaches 95.60%
- Effective assets fall to about 1.09

Empirical finding - not a coding failure.

DEFENSIVE COLLAPSE

HRP minimized drawdown but failed as a return-seeking allocation

1.62%	0.58%	-0.670
Annualized return	Volatility	Sharpe
-0.35%	**7.49%**	**1.09**
Max drawdown	Annualized turnover	Effective assets

POSITIVE

- Very low volatility
- Tiny drawdown
- Low turnover

NEGATIVE

- Return below 2% reference
- Extreme concentration
- Little economic diversification

High Calmar is mechanical: the drawdown denominator is extremely small.

COMMON 159 MONTHS

Performance metrics reveal different objectives, not one universal winner

Strategy	Return	Volatility	Sharpe	Max drawdown
Equal Weight	6.09%	9.11%	0.476	-19.07%
60/40	7.01%	9.76%	0.540	-23.32%
Min Variance Ledoit-Wolf	2.31%	2.88%	0.114	-9.11%
Max Sharpe Ledoit-Wolf	9.86%	8.45%	0.922	-22.02%
Target Volatility Ledoit-Wolf	11.38%	11.63%	0.818	-22.27%
Michaud 500	9.23%	7.98%	0.900	-20.51%
HRP	1.62%	0.58%	-0.670	-0.35%

Target Volatility leads return; Maximum Sharpe leads risk-adjusted performance.

Implementation metrics change how the performance ranking is judged

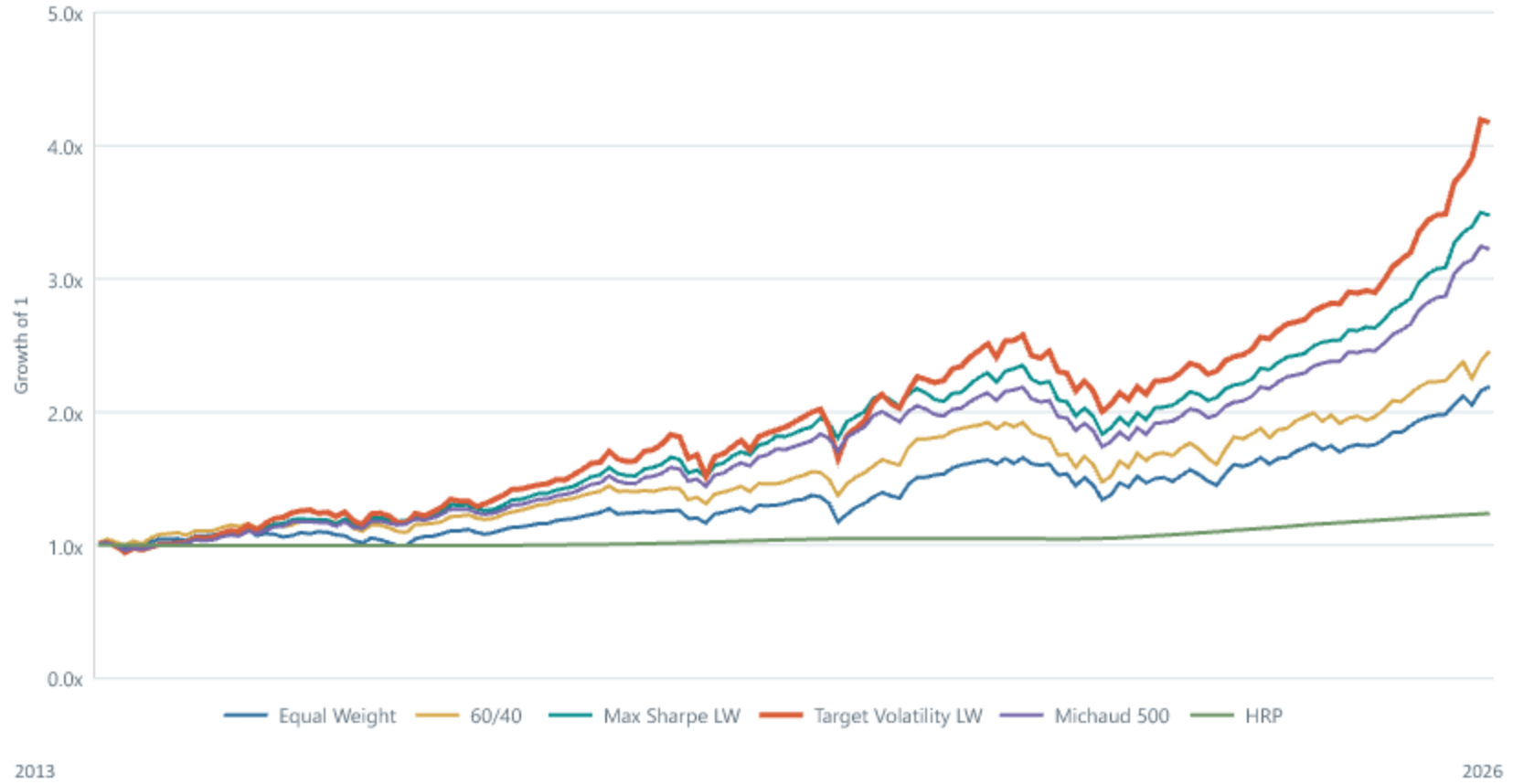
Strategy	Turnover	Effective assets	HHI	Interpretation
Equal Weight	13.23%	23.00	4.35%	Broadest by construction
60/40	12.45%	13.89	7.20%	Simple sleeve allocation
Min Variance Ledoit-Wolf	23.46%	5.35	18.76%	Low-risk defensive mix
Max Sharpe Ledoit-Wolf	88.06%	4.77	21.26%	Higher turnover and concentration
Target Volatility Ledoit-Wolf	98.66%	4.52	22.19%	Highest selected turnover
Michaud 500	78.00%	6.25	16.29%	Broadest optimized allocation
HRP	7.49%	1.09	91.72%	Low turnover because cash dominated

Michaud 500 improves optimized breadth; HRP's low turnover is economically misleading.

SELECTED STRATEGY EQUITY CURVES

FROZEN RETURN SERIES

Terminal wealth reflects both skill and the amount of risk accepted



READ THE PATH

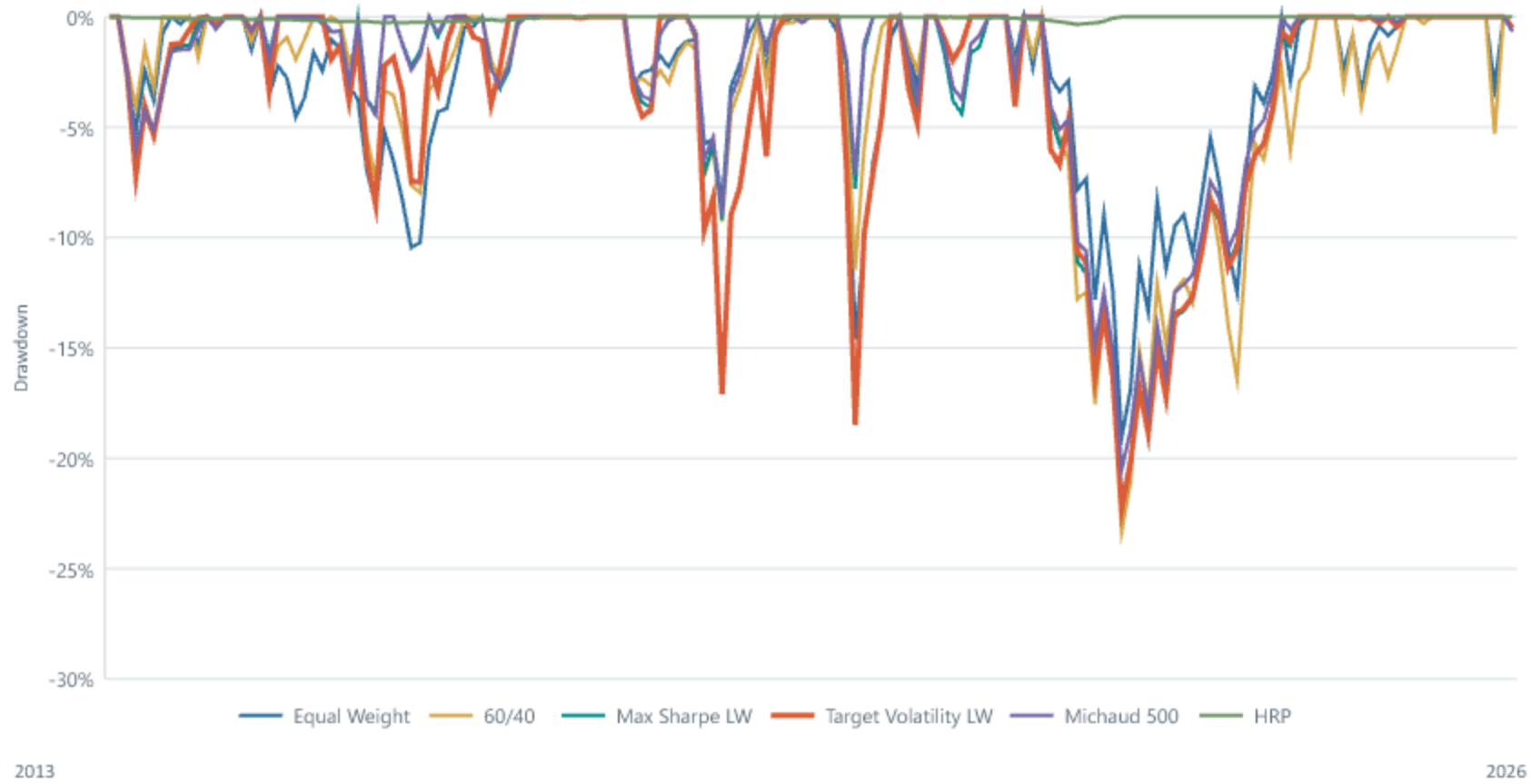
- Target Volatility finishes highest by taking more risk
- Max Sharpe and Michaud deliver strong middle paths
- Benchmarks remain credible but finish below leaders
- HRP behaves like cash after concentrating in BIL

Selected-strategy view; complete 12-strategy chart remains in the report.

SELECTED STRATEGY DRAWDOWNS

LOSS FROM PRIOR PEAK

Drawdowns expose the path risk hidden by terminal wealth



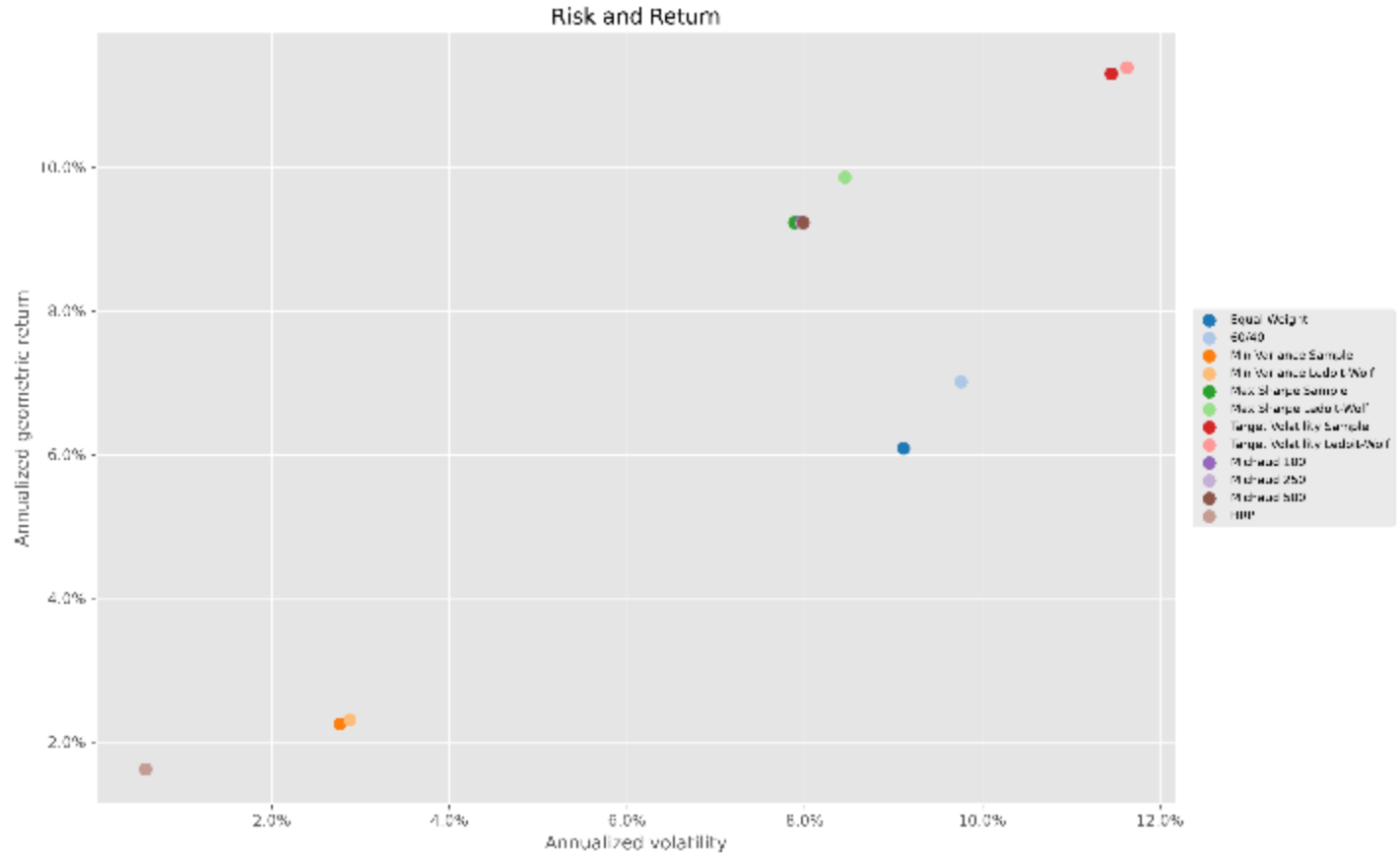
INTERPRETATION

- High wealth still includes material stress
- Minimum Variance and HRP take less risk
- 2022 challenged equity-bond diversification
- Small drawdown alone does not imply high return

Drawdown is path dependent; volatility is not.

RETURN VS VOLATILITY

Maximum Sharpe offered the best realized return-risk balance



Frozen final risk-return output

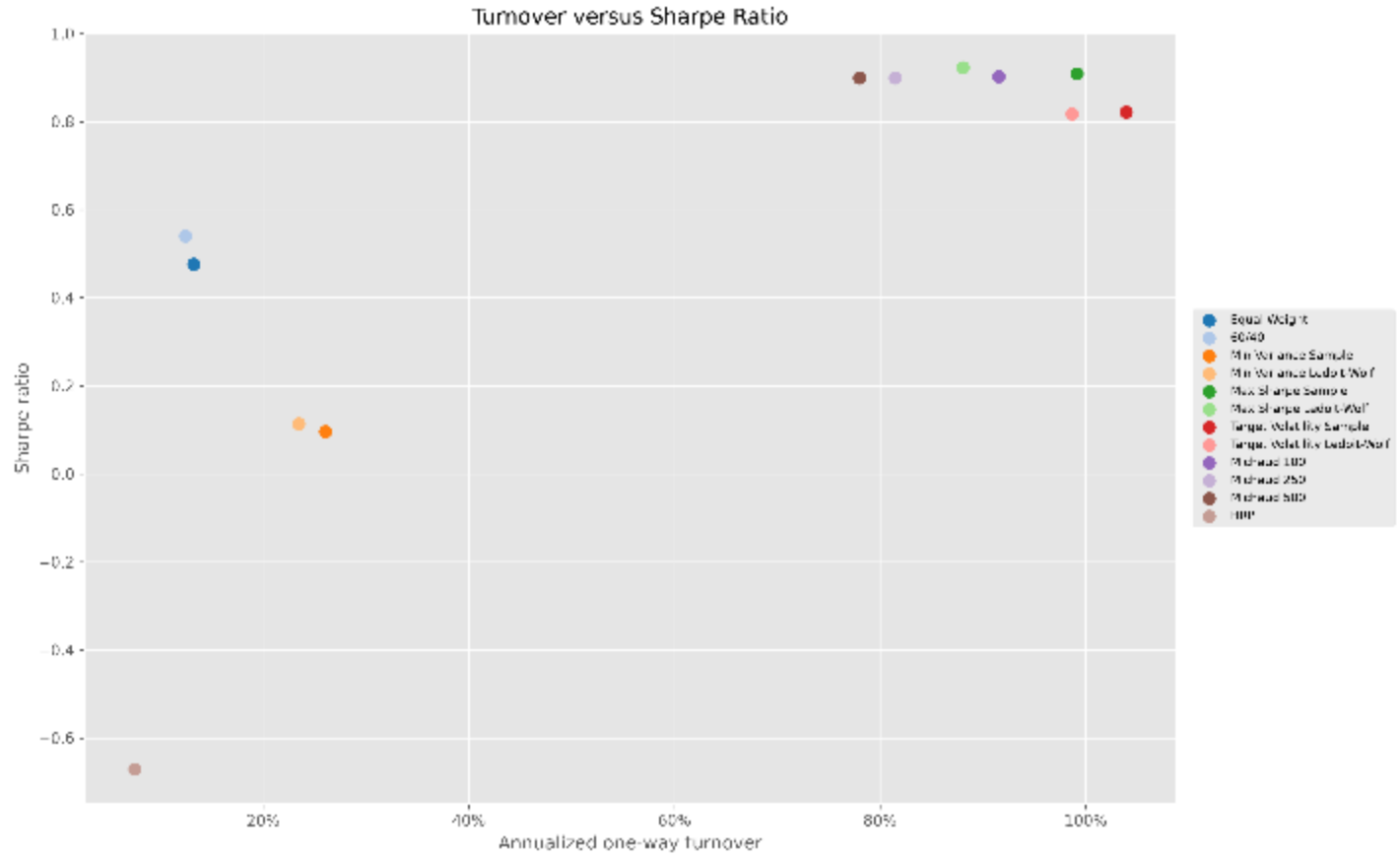
HOW TO READ

- Upper-left is generally preferred
- Target Volatility: more return and more risk
- Maximum Sharpe LW: best realized Sharpe
- Min Variance and HRP sit near the low-risk corner

Risk preference still determines the relevant choice.

IMPLEMENTATION FRONTIER

Higher Sharpe generally required more trading



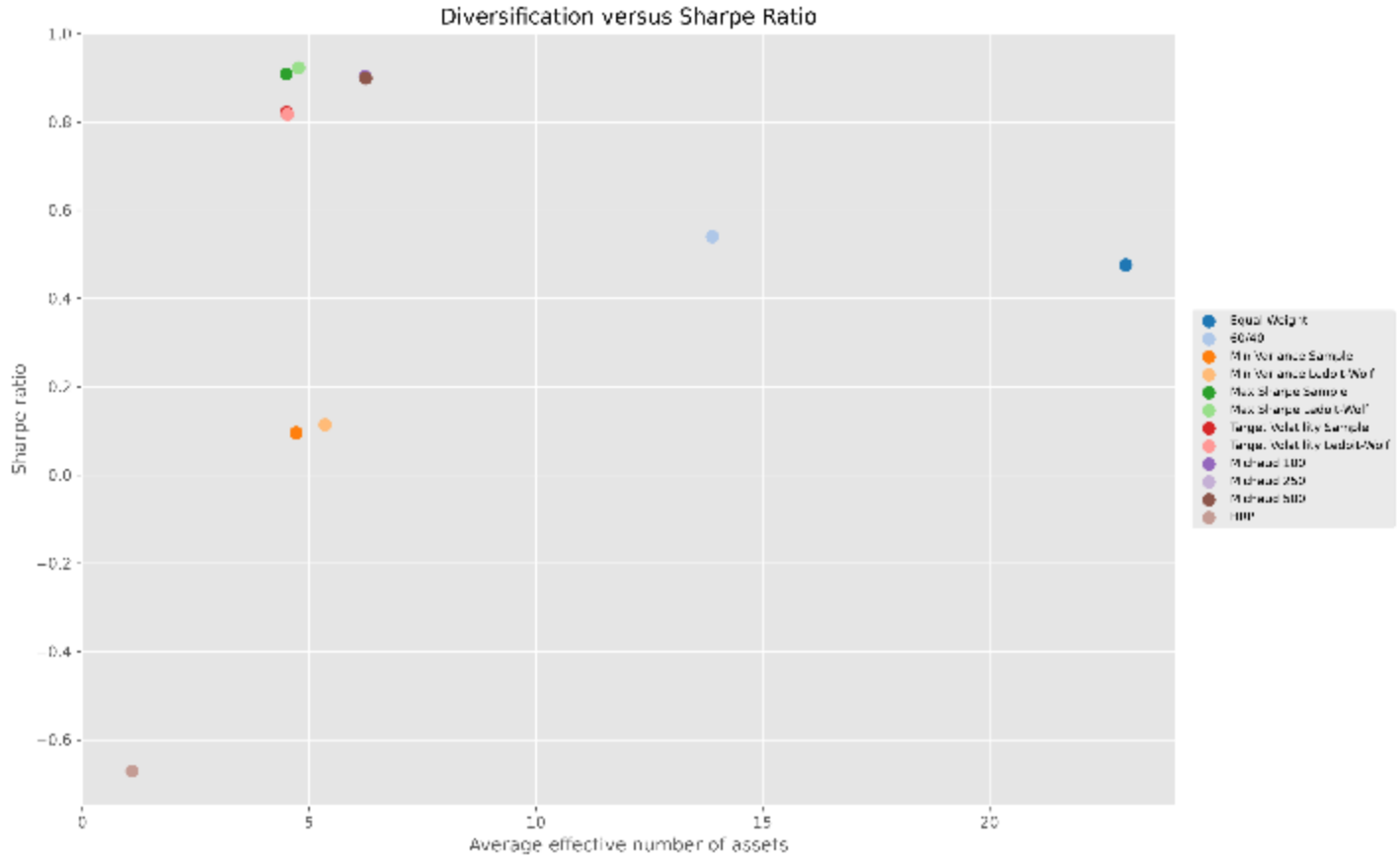
Frozen final turnover-Sharpe output

TRADE-OFF

- Maximum Sharpe led Sharpe but traded heavily
- Michaud kept near-leading Sharpe with less turnover
- Benchmarks were easier to implement
- HRP was stable only because cash dominated

Returns are already net of the common 5-bps cost rule.

Michaud broadened optimized portfolios without losing much Sharpe



Frozen final diversification-Sharpe output

KEY POINTS

- Equal Weight: 23 by construction
- Michaud 500: broadest optimized mix
- Classical optimized portfolios stay concentrated
- HRP is least diversified despite its label

Position count is not effective diversification.

MULTI-DIMENSIONAL ANSWER

The winning strategy depends on the investor's objective

BEST SHARPE

Max Sharpe Ledoit-Wolf

0.922

HIGHEST WEALTH

Target Volatility Ledoit-Wolf

4.17

BEST ROBUSTNESS COMPROMISE

Michaud 500

6.25 effective assets

LOWEST OPTIMIZED RISK

Min Variance Ledoit-Wolf

2.88% volatility

LOWEST TURNOVER

HRP

7.49% - cash dominated

SIMPLEST IMPLEMENTATION

Equal Weight / 60/40

No estimated inputs

No single strategy dominates return, risk, turnover, concentration, and simplicity.

Michaud improved robustness rather than every performance metric

MAXIMUM SHARPE LEDOIT-WOLF

- Uses one estimated market input
- Exploits return signals aggressively
- Achieved the best Sharpe: 0.922
- More concentrated and high-turnover

MICHAUD 500

- Averages decisions across plausible inputs
- Improves allocation breadth
- Reduces annualized turnover to 78.00%
- Sacrifices only 0.022 Sharpe

Resampling moderated the implementation profile; it did not create a universal winner.

Input estimation mattered almost as much as objective choice

NOISY COVARIANCE

Unstable correlations and variances

→

SHRINKAGE

Lower estimation variance

→

BETTER CONDITIONING

More stable risk model

→

REALIZED EFFECT

Lower turnover + broader weights

0.909 -> 0.922

Maximum Sharpe

Realized Sharpe

99.14% -> 88.06%

Maximum Sharpe

Annualized turnover

4.50 -> 4.77

Maximum Sharpe

Effective assets

Ledoit-Wolf stabilizes covariance - not noisy expected returns.

NEGATIVE FINDING = EVIDENCE

HRP shows that avoiding inversion does not eliminate model risk

01

Robustness is specific

Avoiding covariance inversion solves only one problem

02

Universe design matters

A cash-like ETF can dominate variance allocation

03

Constraints still matter

No 25% cap allowed extreme concentration

04

Labels can mislead

Risk parity did not imply diversified capital

The BIL concentration is an empirical method-universe interaction - not a coding error.

Future test: constrained HRP, HRP without cash, and asset-class floors.

LOW MODEL DEPENDENCE

Simple benchmarks remain difficult to dismiss

Transparent

Rules are immediately understandable

Low complexity

No means, covariance, or solver

Lower turnover

Smaller implementation burden

No estimation error

No noisy input forecasts

Reproducible

Easy to rebuild and audit

Optimized leaders improved selected metrics **but required more trading and stronger modeling assumptions**

Benchmarks are economic alternatives - not ceremonial rows in a table.

Added complexity must earn its place after costs.

SCOPE OF CLAIMS

Data and sample choices limit how broadly the evidence can be generalized

Fixed ETF universe

Possible selection and survivorship bias

Yahoo Finance

Not an institutional total-return source

One realized path

No repeated independent market histories

Late inception

Four ETFs shorten full-universe history

Extended 27 panel

Not a complete optimized robustness backtest

The ranking is comparative evidence for this design and period - not universal proof.

TRANSPARENT ASSUMPTIONS

Model assumptions remain simplified and testable

ESTIMATION

- Expected returns remain noisy
- Fixed 60-month estimation window
- Constant 2% risk-free reference
- Michaud normal-sampling assumptions

IMPLEMENTATION

- Constant 5-bps transaction costs
- No taxes or market impact
- No time-varying liquidity model
- No formal significance testing

Transparency makes the experiment reproducible; sensitivity analysis tests how dependent the results are.

PRE-SPECIFY SENSITIVITY

Future robustness checks should challenge the design, not search for winners

WINDOWS

36 / 60 / 120 months

REBALANCING

Monthly vs quarterly

HRP

Cap weights or remove BIL

ASSET CLASSES

Floors and maximum sleeves

BLACK-LITTERMAN

Only with defensible views

REGIMES

Time-varying covariance

COSTS

Asset- and regime-specific

EXTENDED UNIVERSE

Full 27-ETF optimization

Principle: define robustness tests before seeing which specification performs best.

Robust portfolio design is more than choosing an optimizer

01	**Identical out-of-sample months**	Same assets, dates, costs, and next-month timing
02	**Best realized Sharpe**	Maximum Sharpe Ledoit-Wolf: 0.922
03	**Highest terminal wealth**	Target Volatility Ledoit-Wolf: more risk
04	**Best robustness compromise**	Michaud 500: broader weights and lower turnover
05	**Most important negative result**	HRP still depended on constraints and universe design

Sophistication added value only when paired with estimation discipline and implementation controls.

DEFENSE GUIDE

Questions

DATA

Why ETFs?
Why 23?
Why preserve 2008?

DESIGN

Why monthly?
Why 60 months?
Why 5 bps?

METHODS

Why Markowitz?
Why Michaud?
Why Ledoit-Wolf?
Why HRP?

RESULTS

Why did Max Sharpe win?
Why did HRP fail?
Which strategy is practical?

Anchor answer: same universe + same months + same timing + same costs.

Thank you.